Zero-Day Detection under HAI Plant-Version Shift

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The problem

A zero-day detector must flag attacks it has never seen, trained on normal operation alone. The harder case is when the *plant itself* is replaced by a later, differently instrumented version: HAI 21.03 and 23.05 are two such versions of the same testbed, sharing 59 tags but not the instrumentation.

How much does detection degrade from 21.03 to 23.05 — and can an adaptive decision layer recover what is lost?

A detector that works in-domain may stop working when the baseline shifts, and no amount of threshold tuning can repair a representation that no longer separates attacks from normal operation.

Data

partition	files	rows	attacks
21.03 normal training	3	921,603	0
21.03 test	5	402,005	50
23.05 test	2	284,400	52
all streams	10	1,608,008	102

The two versions share 59 public tags; 18 have zero variance throughout the normal 21.03 training data and are removed *using source statistics only*, leaving **41 modelling tags**. Their constant source encoding would otherwise dominate max-fused scores after the shift.

Protocol: no labels, anywhere

- Attack labels are used for **evaluation only** — never for training, model selection, or threshold calibration.
- Normalisation is fitted on chronological 21.03 training partitions; the last 15% supplies normal calibration scores.
- EVT calibrated on **source-normal** scores to an empirical target of **10 false alarms/day**.
- Three seeds (11, 23, 37). No point adjustment.

References

[1] H.-K. Shin et al. *HAI: HIL-based Augmented ICS Security Dataset.* CSET 2020.
[2] L. Ruff et al. *Deep One-Class Classification.* ICML 2018.
[3] L. Ruff et al. *Rethinking Assumptions in Deep Anomaly Detection.* arXiv:2006.00339, 2020.
[4] A. Siffer et al. *Anomaly Detection in Streams with Extreme Value Theory.* KDD 2017.
[5] W.-S. Hwang et al. *Time-Series Aware Precision and Recall for Anomaly Detection.* CIKM 2019.
[6] W.-S. Hwang et al. *Do You Know Existing Accuracy Metrics Overrate Time-Series Anomaly Detections?* SAC 2022.
[7] S. Kim et al. *Towards a Rigorous Evaluation of Time-series Anomaly Detection.* AAAI 2022.

What we built

Three detectors, trained on normal 21.03 data only:

- **PCA** reconstruction on flattened windows.
- **Deep SVDD** — a three-layer bias-free 1-D convolutional trunk into a latent hypersphere (13,904 parameters).
- **Convolutional VAE** — the same trunk, plus Gaussian heads and a decoder (26,352 parameters).

Windows are causal (60 s, scored at the window end, 10 s evaluation stride). Per-sensor scores are fused by **maximum**, so an alarm needs one strongly anomalous sensor. The two neural models share the trunk but not the total parameter count, and that gap is reported rather than hidden.

Result 1: in-domain, at a realistic alarm budget

At 10 false alarms per day on 21.03, all three detectors work moderately — and none of them works well:

	event recall	eTaF1	PR-AUC	Hit@3
PCA	0.740	0.533	0.502	0.229
Deep SVDD	**0.773**	**0.543**	0.489	0.171
VAE	0.560	0.473	0.462	**0.257**

A PR-AUC near 0.5 at a 1–2% attack rate is a weak ranker, not a solved problem. That matters for what follows: there was little to lose.

Result 2: the representation collapses

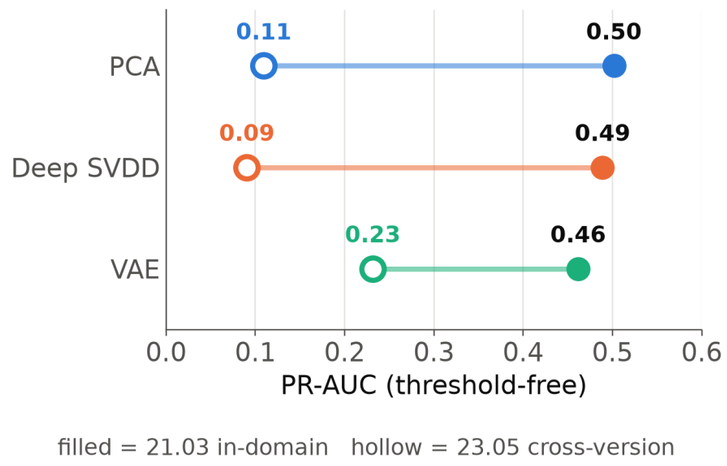


Figure 1: PR-AUC is threshold-free, so it measures the representation alone.

Every model loses most of its ranking quality: PCA 0.50 $\rightarrow$ 0.11, Deep SVDD 0.49 $\rightarrow$ 0.09, VAE 0.46 $\rightarrow$ 0.23. VAE retains the most; PCA and Deep SVDD retain almost nothing. This is the failure that a decision layer cannot repair.

Result 3: high recall can be an alarm collapse

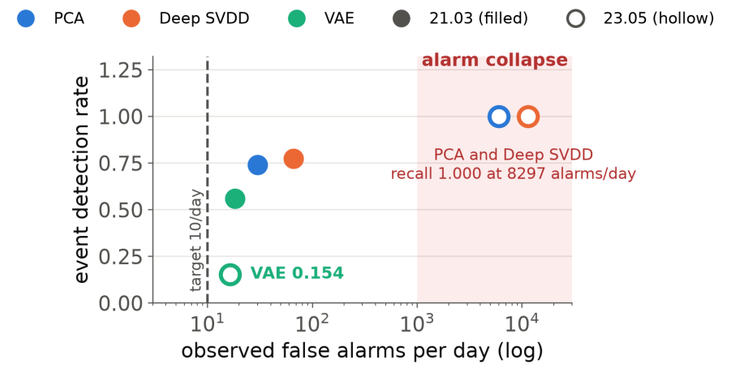


Figure 2: Frozen arm. The calibration target was 10 alarms/day.

PCA and Deep SVDD report **event recall 1.000** on 23.05 — because they fire **8,297 alarms per day**, on 96% of normal windows. A circular-shift null settles it: a randomly time-shifted version of the same alarm series matches this recall in every one of 200 shifts ($p = 1.00$). On 21.03 the same models beat their null at $p < 0.001$. VAE reports the honest number: recall 0.154 at 16.4 alarms/day, still 7.7$\times$ its null. Raw recall is not a standalone measure of useful detection.

Result 4: DSPOT does not recover it

model	frozen	adaptive	gain	FA/day
	23.05 event recall			*frozen→adapt.*
PCA	1.000†	1.000†	0.000	8297→8297
Deep SVDD	1.000†	1.000†	0.000	8297→8297
VAE	0.154	0.115	**−0.038**	16.4→5.5

†Recall 1.000 with 8,297 alarms/day is *alarm collapse*: a circular-shift null reaches 1.000 in 200 of 200 shifts ($p = 1.00$). It is not detection.

For VAE, DSPOT cuts alarms from 16.4 to 5.5 per day but *loses* recall (recovery ratio −0.095). For the other two, nothing changes: they were already alarmed everywhere.

Why: PR-AUC is *identical* between the frozen and adaptive arms — 4.873e-05 vs 9.481e-05 for the EVT risk parameter, same scores. DSPOT moves the decision threshold; it does not touch the encoder, the PCA basis, or the anomaly score. **Decision-layer adaptation cannot substitute for representation quality.**

Conclusions

- **Zero-day detection does not survive the plant-version shift.** Ranking quality falls by roughly a factor of four to five; two of three detectors collapse into continuous alarm.
- **The decision layer is not the bottleneck.** DSPOT reduces alarms where the representation still works and does nothing where it does not.
- **Report threshold-free metrics alongside event recall.** On the shifted domain, recall 1.000 and recall 0.154 can describe equally useless detectors.
- Limits: one testbed, two versions, three seeds; the source-normal calibration targets an empirical source rate, not a guaranteed shifted-domain budget.